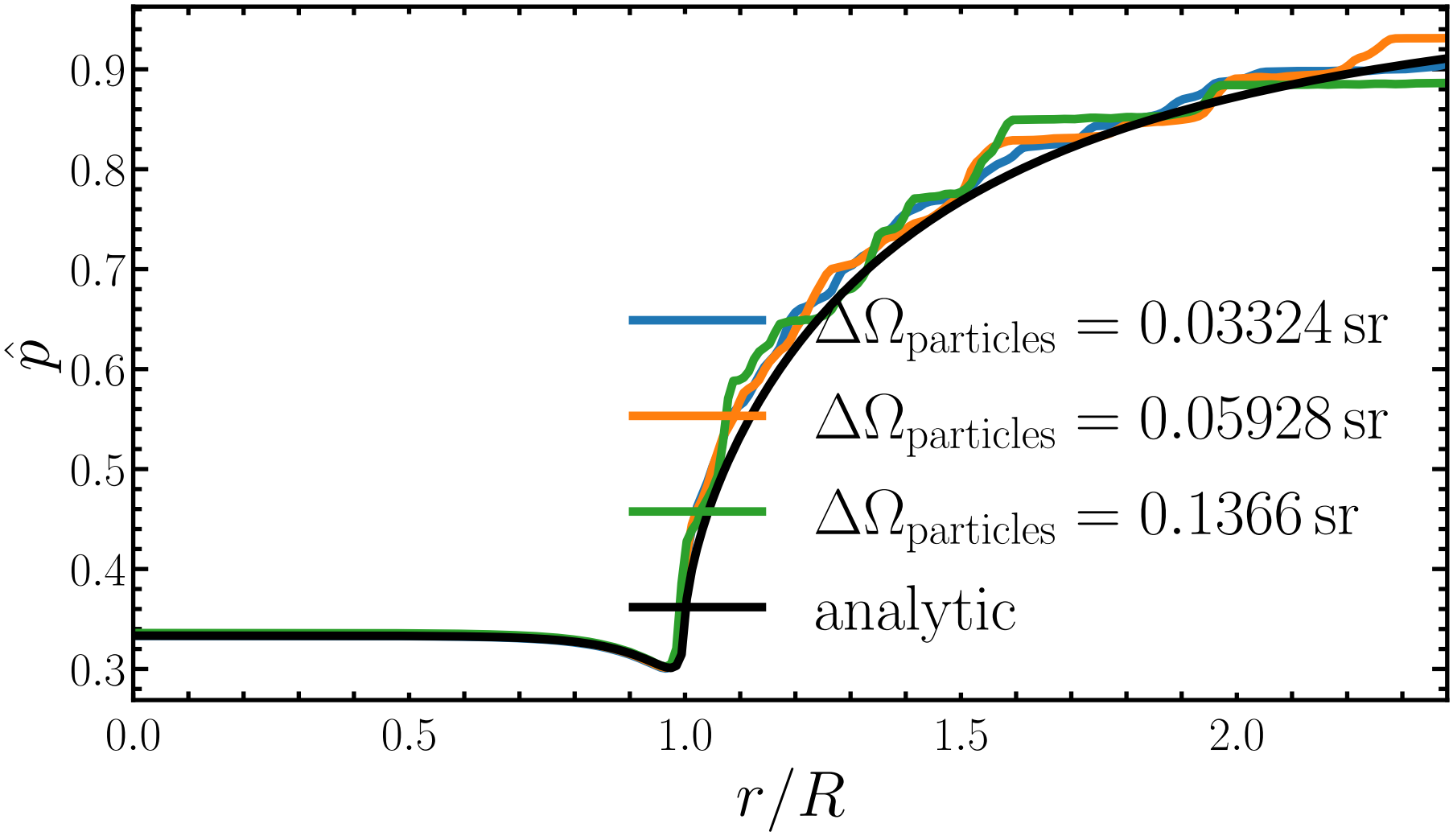

0.9
0.8
0.7
0.6
0.5
0.4
0.3
$\hat{p}$
0.0
0.5
1.0
1.5
2.0
r/R
$\Delta\Omega_{\mathrm{particles}} = 0.03324\,\mathrm{sr}$
$\Delta\Omega_{\mathrm{particles}} = 0.05928\,\mathrm{sr}$
$\Delta\Omega_{\mathrm{particles}} = 0.1366\,\mathrm{sr}$
analytic